Module 1 — Overview of OpenStack

Cloud computing basics, service/deployment models and OpenStack logical architecture

Answer-length target

4 marks: prepare about **2 pages** in your 19 × 15.5 cm book. 8 marks: prepare about **3–4 pages**. The bank contains practice questions, not model answers.

Section A — 4-mark questions

For short answers, prepare a clear definition, key features/components, and a small example or diagram when relevant.

Q1. Define cloud computing. State any four advantages of cloud computing. **[HIGH]**

Q2. Differentiate public cloud, private cloud and hybrid cloud with one example each. **[HIGH • PYQ]**

Q3. Compare IaaS, PaaS and SaaS. **[HIGH • PYQ-aligned]**

Q4. What is OpenStack? State its purpose in cloud computing. **[HIGH]**

Q5. Explain the role of Keystone in OpenStack. **[HIGH]**

Q6. Write a short note on Swift object storage and its benefits. **[MEDIUM]**

Q7. What is Cinder? List its main volume-management features. **[MEDIUM]**

Q8. Explain the purpose of the OpenStack Heat service. **[HIGH • PYQ]**

Q9. What is the role of Horizon in OpenStack? **[HIGH]**

Q10. Explain the role of the message queue in OpenStack architecture. **[MEDIUM]**

Section B — 8-mark questions

For long answers, prepare an introduction, labelled diagram where relevant, components/working, comparison or advantages, and a short conclusion.

Q1. Compare the different components in OpenStack logical architecture with a neat diagram. **[HIGH • PYQ]**

Q2. Describe and compare the three main cloud service models. Explain how they support cloud computing. **[HIGH • PYQ]**

Q3. Explain the public, private and hybrid cloud deployment models with characteristics and examples. **[HIGH]**

Q4. Explain the purpose of OpenStack API services and how they enable interaction among OpenStack components. **[HIGH • PYQ]**

Q5. Discuss the importance and key functionalities of the OpenStack Horizon dashboard for users and administrators. **[HIGH • PYQ]**

Q6. Compare block, object and file-share storage in OpenStack. Give a use case for each. **[HIGH • PYQ]**

Q7. Describe the architecture of OpenStack Swift and explain how it stores and retrieves unstructured data. **[HIGH • PYQ]**

Q8. Explain the major OpenStack services—Keystone, Nova, Glance, Neutron and Horizon—and how they work together. **[HIGH]**

Q9. Explain OpenStack Heat orchestration, its main features and the use of templates/stacks. **[MEDIUM]**

Q10. Explain the role of the database and message queue in the OpenStack logical architecture. **[MEDIUM]**

Priority key: HIGH = major topic in the module notes and/or supported by the previous-year paper. MEDIUM = supporting topic worth revising after the high-priority list.

Question priorities use Module 1 notes and the uploaded October 2025 paper. Questions are practice prompts, not guaranteed predictions.

Module 2 — OpenStack Cluster, Cloud Controller & Common Services

Clustering, Keystone identity, federation, Fernet tokens and controller services

Answer-length target

4 marks: prepare about **2 pages** in your 19 × 15.5 cm book. 8 marks: prepare about **3–4 pages**. The bank contains practice questions, not model answers.

Section A — 4-mark questions

For short answers, prepare a clear definition, key features/components, and a small example or diagram when relevant.

Q1. Define server clustering and explain vertical scaling and horizontal scaling. **[HIGH]**

Q2. Explain asymmetric clustering. **[HIGH]**

Q3. Explain symmetric clustering. **[HIGH]**

Q4. Differentiate symmetric and asymmetric clustering. **[HIGH • PYQ]**

Q5. List the main responsibilities of the OpenStack cloud controller. **[HIGH]**

Q6. Explain the role of Keystone in identity management and service cataloguing. **[HIGH]**

Q7. What is federated identity? State any four advantages. **[HIGH]**

Q8. What are Fernet tokens? State their main advantages. **[HIGH]**

Q9. Differentiate Nova conductor and Nova scheduler. **[HIGH • PYQ]**

Q10. Write a short note on the Horizon dashboard or Nova API service. **[MEDIUM]**

Section B — 8-mark questions

For long answers, prepare an introduction, labelled diagram where relevant, components/working, comparison or advantages, and a short conclusion.

Q1. Discuss Keystone identity management in OpenStack, explaining its main providers. **[HIGH • PYQ]**

Q2. Describe the functionality of Federated Keystone and Fernet tokens. **[HIGH • PYQ]**

Q3. Differentiate Nova conductor and Nova scheduler services in detail. **[HIGH • PYQ]**

Q4. Explain the cloud controller and the services it provides in an OpenStack deployment. **[HIGH]**

Q5. Explain symmetric and asymmetric clustering, comparing availability, workload distribution and failure handling. **[HIGH]**

Q6. Explain the federated identity authentication process and its advantages. **[HIGH]**

Q7. Explain Fernet tokens, their contents, benefits and key-rotation requirement. **[HIGH]**

Q8. Describe Nova scheduler filtering and weighting and explain how it chooses a compute host. **[MEDIUM]**

Q9. Explain the role of Nova API and Glance image management in a VM request. **[MEDIUM]**

Q10. Explain how the controller supports APIs, messaging, database services and high availability. **[MEDIUM]**

Priority key: HIGH = major topic in the module notes and/or supported by the previous-year paper. MEDIUM = supporting topic worth revising after the high-priority list.

Questions are based on Module 2 notes and the uploaded October 2025 Cloud Computing paper.

Module 3 — OpenStack Compute & Storage

Nova components, hypervisors, containers, Magnum, storage and compute-resource segregation

Answer-length target

4 marks: prepare about **2 pages** in your 19 × 15.5 cm book. 8 marks: prepare about **3–4 pages**. The bank contains practice questions, not model answers.

Section A — 4-mark questions

For short answers, prepare a clear definition, key features/components, and a small example or diagram when relevant.

Q1. List the main components of the OpenStack compute service and state their roles. **[HIGH • PYQ-aligned]**

Q2. Differentiate Nova conductor and Nova scheduler. **[HIGH • PYQ]**

Q3. Define a hypervisor. Name hypervisors supported in the notes. **[MEDIUM]**

Q4. Differentiate para-virtualization and complete virtualization. **[MEDIUM]**

Q5. What is a Docker container? State its advantages. **[MEDIUM]**

Q6. What is the OpenStack Magnum project? **[HIGH]**

Q7. Define Availability Zone and Host Aggregate. **[HIGH]**

Q8. Differentiate ephemeral and persistent storage. **[HIGH]**

Q9. What is CPU/RAM overcommitment? Define a flavor. **[MEDIUM]**

Q10. List the main steps involved in launching an OpenStack instance. **[HIGH]**

Section B — 8-mark questions

For long answers, prepare an introduction, labelled diagram where relevant, components/working, comparison or advantages, and a short conclusion.

Q1. Describe the primary components of the OpenStack Compute service and their roles in managing virtual instances. **[HIGH • PYQ]**

Q2. Explain the Nova scheduling process and the steps involved in booting a virtual machine. **[HIGH]**

Q3. Compare Swift, Cinder and Manila, explaining the purpose and use case of each storage service. **[HIGH • PYQ]**

Q4. Describe the Swift ring with a neat diagram. **[HIGH • PYQ]**

Q5. Explain the OpenStack Magnum project, its architecture, components and working. **[HIGH]**

Q6. Explain availability zones, host aggregates, Nova cells and regions as methods of segregating compute resources. **[HIGH]**

Q7. Explain the hypervisor and compare virtualization approaches discussed in the notes. **[MEDIUM]**

Q8. Discuss external shared and internal non-shared instance storage, including advantages and drawbacks. **[MEDIUM]**

Q9. Explain CPU and RAM allocation ratios, overcommitment and the role of flavors. **[MEDIUM]**

Q10. Explain how Docker containers work and compare their characteristics with virtual-machine-based deployment. **[MEDIUM]**

Priority key: HIGH = major topic in the module notes and/or supported by the previous-year paper. MEDIUM = supporting topic worth revising after the high-priority list.

Module 3 priorities reflect the uploaded slides and related questions in the October 2025 paper.

Module 4 — Neutron & OpenStack Networking

Neutron architecture, plugins, virtual networks, routers, floating IPs and network security

Answer-length target

4 marks: prepare about **2 pages** in your 19 × 15.5 cm book. 8 marks: prepare about **3–4 pages**. The bank contains practice questions, not model answers.

Section A — 4-mark questions

For short answers, prepare a clear definition, key features/components, and a small example or diagram when relevant.

Q1. What is Neutron? List its main functions. **[HIGH]**

Q2. List and explain the main components of Neutron architecture. **[HIGH]**

Q3. Explain the ML2 plugin and its significance in implementing virtual networks. **[HIGH • PYQ]**

Q4. Differentiate VLAN-based networks and tunnel-based networks. **[HIGH • PYQ]**

Q5. Define network, subnet and port in Neutron. **[MEDIUM]**

Q6. What is a floating IP? State its purpose. **[HIGH]**

Q7. Explain the role of a virtual router and external gateway. **[HIGH]**

Q8. What are security groups? **[HIGH]**

Q9. List the functions of L2, DHCP, L3 and VPN agents. **[MEDIUM]**

Q10. What is Firewall as a Service (FWaaS)? **[MEDIUM]**

Section B — 8-mark questions

For long answers, prepare an introduction, labelled diagram where relevant, components/working, comparison or advantages, and a short conclusion.

Q1. Explain the architecture of the OpenStack Neutron service and how its components provide networking to virtual instances. **[HIGH • PYQ]**

Q2. Distinguish VLAN-based and tunnel-based networks, explaining how they isolate tenant traffic. **[HIGH • PYQ]**

Q3. Explain the role of floating IPs and how they enable external connectivity to virtual machines. **[HIGH • PYQ]**

Q4. Describe Neutron's core plugin, service plugins, agents and API extensions. **[HIGH]**

Q5. Explain how virtual networks are connected using routers and how external connectivity is provided. **[HIGH]**

Q6. Explain the ML2 plugin, network types and virtual switches. **[HIGH]**

Q7. Discuss security groups and Firewall as a Service, including policies and rules. **[HIGH]**

Q8. Explain Neutron subnets, CIDR, gateways, DHCP and allocation pools. **[MEDIUM]**

Q9. Describe the functions of the L2, DHCP, L3 and VPN agents. **[MEDIUM]**

Q10. Explain the steps for associating a floating IP with a VM and discuss the role of security rules. **[HIGH]**

Priority key: HIGH = major topic in the module notes and/or supported by the previous-year paper. MEDIUM = supporting topic worth revising after the high-priority list.

Module 4 questions are based on the uploaded Neutron notes and topics explicitly asked in the October 2025 paper.

Module 5 — Heat, High Availability & Terraform

Orchestration, Heat templates/stacks, high availability, database/message-queue HA and RabbitMQ clustering

Answer-length target

4 marks: prepare about **2 pages** in your 19 × 15.5 cm book. 8 marks: prepare about **3–4 pages**. The bank contains practice questions, not model answers.

Section A — 4-mark questions

For short answers, prepare a clear definition, key features/components, and a small example or diagram when relevant.

Q1. Define cloud orchestration and state its main purposes. **[HIGH]**

Q2. What is OpenStack Heat? List its main functions. **[HIGH]**

Q3. Define a Heat stack and a Heat template. **[HIGH]**

Q4. List the main components of OpenStack Heat. **[HIGH]**

Q5. What is High Availability (HA)? State its main objective. **[HIGH]**

Q6. Define database high availability and explain replication briefly. **[HIGH]**

Q7. What is message-queue high availability and why is it needed in OpenStack? **[HIGH]**

Q8. What is RabbitMQ? Explain RabbitMQ clustering. **[HIGH]**

Q9. What is Terraform? Define Infrastructure as Code (IaC). **[HIGH]**

Q10. State the main features of Terraform. **[MEDIUM]**

Section B — 8-mark questions

For long answers, prepare an introduction, labelled diagram where relevant, components/working, comparison or advantages, and a short conclusion.

Q1. Explain the architecture of OpenStack Heat and describe its main components and functions. **[HIGH]**

Q2. Describe the process of creating and managing a Heat stack, including template submission and resource creation. **[HIGH]**

Q3. Explain the Heat stack lifecycle: creation, update and deletion. **[HIGH]**

Q4. Discuss High Availability in OpenStack and explain the mechanisms used to reduce single points of failure. **[HIGH]**

Q5. Explain database HA architecture and the role of replication and failover. **[HIGH]**

Q6. Explain high availability in a message queue and compare normal and HA message-queue setups. **[HIGH]**

Q7. Explain RabbitMQ clustering architecture and how it supports reliable communication between OpenStack services. **[HIGH]**

Q8. Explain Terraform-based orchestration in OpenStack and outline its workflow. **[HIGH]**

Q9. Compare OpenStack Heat and Terraform for infrastructure orchestration, using only the features described in the notes. **[MEDIUM]**

Q10. Explain how redundancy, failover, load balancing and database replication maintain service availability. **[HIGH]**

Priority key: HIGH = major topic in the module notes and/or supported by the previous-year paper. MEDIUM = supporting topic worth revising after the high-priority list.

Module 5 topics were extracted by reading the image-based slides in the uploaded CC MOD 5.pptx. The slides cover Heat/orchestration, HA, database and message-queue HA, Terraform and RabbitMQ clustering.